

Mahir Gandhi

East Lansing, MI | 463-254-9796 | gandhim3@msu.edu | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Michigan State University, College of Engineering, Honors College

East Lansing, MI

Bachelor of Science, Mechanical Engineering – Manufacturing Concentration (GPA: 3.9/4.0)

Expected 2027

- **Highlights:** Entrepreneurship minor, Resident Assistant, College Mathematics Tutor, Michigan EV Scholar, Dean's List (5X), 1st Place—MSU 2 Day Venture Challenge, Best Accessible & Inclusive Design—MSU Designathon.
- **Certifications & Trainings:** Lean Six Sigma Green Belt (AIGPE), Minitab (AIGPE), Project Management (PMI), GD&T (Geomiq), Siemens NX (Tata Technologies), SOLIDWORKS (Tesla), FMEA (General Motors), Mill & Lathe (MSU)

EXPERIENCE

Zoox, Inc.

May 2026 – Aug 2026

Supplier Quality Engineer Intern - Zoox Autonomous Robotaxi

Hayward, CA

- Resolved a critical PCU air-leak defect through DOE, poka-yoke fixture design, PFMEA, assembly-line budgeting and operator training, reducing repeat defects **80%**
- Closed PPAP nonconformances by resolving **3** seat defects across **2** lamination and injection molding suppliers
- Built Zoox's first cross-platform SQE dashboard using Next.js, React, and Jira/JQL, centralizing supplier defects, corrective-action status, and performance tracking across **30+** suppliers with **50+** concurrent users

Tesla, Inc.

August 2025 – December 2025

Quality Engineer Intern - Model S, Model X

Fremont, CA

- Designed and integrated electromechanical protective equipment for large-scale alignment systems (SolidWorks + CATIA 3DX), incorporating fail-safes and error-proofing to reduce unplanned maintenance
- Drove new product validation for hinges through design & supplier coordination, driving **18%** torque improvement
- Combined time studies with **5+** metrology and process datasets to execute 8D/RCA trials for door-closing effort defects, reducing defects over **30%** and ending a year-long containment
- Diagnosed leak paths and failure points on **20+** trunk panel water leak defects, implementing process containments and driving up to a **100%** reduction on certain repeat defects through data-driven root-causing and DOE
- Directed crossed gage R&R in Minitab for a door-assembly station, isolating measurement variation in trial data

General Motors

May 2025 – August 2025

Dimensional Engineer Intern - GMC Acadia, Buick Enclave, Chevy Traverse

Lansing, MI

- Designed, fabricated and implemented **6** 3D-printed fixtures and tools (Siemens NX + Teamcenter) with PFMEA, reducing equipment weight **70%**, saving **\$10,000+** in maintenance costs and improving repeatability across **3** processes
- Initiated **20+** data-driven tooling moves and troubleshoot **10+** FANUC automation faults, reducing defect call-outs **15%**
- Applied Lean Six Sigma methodologies to lead practical problem solving in internal audits, root-causing **50+** findings

Physical Ultrasonics, Microscopy & Acoustics (P.U.M.A) Lab

May 2024 – Present

Honors Research Scholar, Research Intern - Advanced Materials, Manufacturing Processes

East Lansing, MI

- Ran DOE for **30+** ultrasonically deformed copper, mapping **5** process variables to material-response trends.
- Developed novel characterization processes for **15** 3D-printed composites using precision metallurgy and microscopy
- Led studies on polymer ultrasonic welding as a repair technology, achieving **45%** micro-crack closure on damaged samples

STUDENT LEADERSHIP

Michigan State University Solar Racing Team

September 2023 – May 2026

Motor Housing Lead - Cynisca

East Lansing, MI

- Custom-designed **12+** motor housing components, reduced overall design time by **25%** via rigorous weekly design reviews
- Assembled the electromechanical motor stack, engineered the mechanical test fixture and led back-EMF motor testing
- Machined **15+** components using GD&T-compliant engineering drawings to a ± 0.003 -inch tolerance across **100+** hours
- Validated components and assemblies through static structural FEA, driving a **6%** housing weight reduction

Spartan Technical Consulting Group

May 2026 – Present

Founder & Co-President

East Lansing, MI

- Founded MSU's first engineering consultancy, securing **2 industry partnerships** for student-led engineering projects
- Led **8 consultants and board members**, owning sourcing, project scoping, resource allocation, and client delivery

NASA Lspace Program

January 2025 – May 2025

Lead Thermal Engineer - Orpheus

East Lansing, MI

- Owned thermal architecture, CAD (SiemensNX), BOM control and DFMEA for a Venus aerobot, reducing mass by **12%**

SKILLS

Design & Manufacturing: CAD (Siemens NX, SOLIDWORKS, CATIA V5), PLM (Teamcenter, 3DX), ANSYS FEA, GD&T, DFM/DFA, Lean Six Sigma tools (8D/RCA, FMEA, DOE), Fabrication, machining, 3D printing, ultrasonic welding

Process & Data: Microsoft Office 365, Python, Power BI, SQL, Jira/JQL, Minitab, APQP/PPAP, PFMEA